

Artificial Intelligence and Neural Networks

**Fine-tuning LLM for Russian Business Letter
Generation**

Variant 12

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1 Introduction

This report presents the implementation of Variant 12: fine-tuning a Russian-language large language model (LLM) for the task of generating business letter replies. Given an incoming letter and a style instruction, the model produces a formal response in the Russian business correspondence style.

1.1 Task Summary

- **Base model:** `ai-forever/rugpt3large_based_on_gpt2` (RuGPT-3 Large, 760M parameters)
- **Fine-tuning method:** LoRA (Low-Rank Adaptation) via `transformers + peft`
- **Dataset:** 5,500 synthetically generated Alpaca-style business letter pairs
- **Evaluation:** 30 test examples scored 1–5 on three axes; inference speed benchmarked
- **Demo:** Streamlit web application (`app.py`)

1.2 Tools and Libraries

Library	Version / Role
<code>transformers</code>	Model loading, training, inference
<code>peft</code>	LoRA adapter implementation
<code>datasets</code>	Dataset loading
<code>torch</code>	Deep learning backend (Apple MPS)
<code>streamlit</code>	Interactive demo UI

2 Dataset

2.1 Generation Methodology

The dataset was generated synthetically using Python templates. Each example consists of an *incoming letter* and a corresponding *reply*, covering realistic Russian business correspondence scenarios.

2.2 Format

The dataset follows the Alpaca instruction-tuning format:

```
{
  "instruction": "Составьте" вежливый профессиональный ответ...",
  "input": "Уважаемые" коллеги, \Прошун предоставить...",
  "output": "Уважаемый(ая)" (-) АВ.. Петрова, \n\Благодарим...",
  "category": "запрос"информации_",
  "style": "вежливое""
}
```

2.3 Splits

Split	Examples	Purpose
Train	5,000	Weight updates
Validation	250	Monitor overfitting
Test	250	Final evaluation (held-out)
Total	5,500	

2.4 Categories and Styles

Five letter categories were generated with equal distribution:

Category	Russian name
Information Request	zaprosh_informatsii
Complaint	zhaloba
Commercial Proposal	kommercheskoe_predlozhenie
Notification	uvedomlenie
Approval	soglasovanie

Each category has three response styles: **polite** (вежливое), **concise** (краткое), and **assertive** (настойчивое). Total combinations: $5 \times 3 = 15$.

2.5 Tokenization Statistics

Split	Total tokens
Train	843,784
Validation	41,668

3 Architecture and Methodology

3.1 Base Model: RuGPT-3 Large

RuGPT-3 Large is a GPT-2 architecture model pre-trained on Russian text by Sber AI. Key properties:

- **Parameters:** 760M total
- **Architecture:** 24 transformer layers, hidden size 1536, 24 attention heads
- **Vocabulary:** 50,257 BPE tokens
- **Context length:** 2,048 tokens
- **Attention projection:** single `c_attn` Conv1D (1536 $\rightarrow$ 4608, i.e. Q/K/V combined)

3.2 LoRA: Low-Rank Adaptation

LoRA [1] freezes the pre-trained model weights and injects trainable rank-decomposition matrices. For each weight matrix $W \in \mathbb{R}^{d \times k}$:

$$W' = W + \Delta W = W + B \cdot A, \quad A \in \mathbb{R}^{r \times k}, B \in \mathbb{R}^{d \times r}$$

The output is scaled by α/r , where α is a hyperparameter:

$$h = xW^T + \frac{\alpha}{r} \cdot xA^TB^T$$

Only matrices A and B are trained. With $r = 8$ and $d = 1536$, $k = 4608$, each LoRA module adds $8 \times 1536 + 8 \times 4608 = 49,152$ parameters. Across 24 layers: $24 \times 49,152 \approx 1.18\text{M}$ trainable parameters.

3.3 LoRA Configuration

Parameter	Value	Description
<code>lora_r</code>	8	Rank of decomposition matrices
<code>lora_alpha</code>	16	Scale factor; effective scale = $\alpha/r = 2.0$
<code>lora_dropout</code>	0.05	Dropout on LoRA layers
<code>target_modules</code>	<code>c_attn</code>	Applied to combined Q/K/V projection
Trainable parameters	1,179,648	0.15% of 761M total
Adapter file size	4.5 MB	vs ~ 1.5 GB for full model

3.4 Why LoRA FP16, Not QLoRA

QLoRA [2] adds 4-bit NF4 quantization via the `bitsandbytes` library. This requires NVIDIA CUDA. Our machine runs **Apple MPS** (Metal Performance Shaders), which is not supported by `bitsandbytes`. Therefore we used LoRA with FP16 precision. For RuGPT-3 Large (760M), FP16 is memory-efficient enough without quantization.

4 Training

4.1 Prompt Template

Each training example is formatted as:

```
### Инструкция:
{instruction}

### Входящее письмо:
{input}

### Ответ:
{output}
```

4.2 Label Masking

The cross-entropy loss is computed only on the *reply* tokens, not on the instruction and input prefix. This is done by setting `labels[:prefix_len] = -100`, where `-100` is PyTorch's ignore index. Without masking, the model would waste capacity memorizing the prompts.

4.3 Training Hyperparameters

Parameter	Value	Notes
<code>num_train_epochs</code>	3	Full passes over 5,000 examples
<code>learning_rate</code>	2×10^{-4}	Standard for LoRA fine-tuning
<code>per_device_batch_size</code>	4	Per MPS device
<code>gradient_accumulation_steps</code>	4	Effective batch = $4 \times 4 = 16$
<code>lr_scheduler</code>	cosine	Smooth LR decay
<code>warmup_ratio</code>	0.03	3% of steps for LR warmup
<code>fp16</code>	True	Half precision
<code>max_seq_length</code>	256	Truncation limit
<code>optimizer</code>	AdamW	Default
<code>seed</code>	42	Reproducibility

4.4 Training Outcome

- **Training time:** 71.7 minutes (Apple MPS)
- **Checkpoints saved:** epochs 1, 2, 3 (steps 313, 626, 939)
- **Best model:** loaded at end of training (lowest eval loss)
- **Adapter saved to:** `./lora_adapter/` (4.5 MB `.safetensors`)

5 Inference

5.1 Generation Configuration

Parameter	Value	Effect
<code>max_new_tokens</code>	200	Maximum reply length
<code>do_sample</code>	True	Stochastic sampling
<code>temperature</code>	0.7	Sharpens distribution (< 1 = less random)
<code>top_p</code>	0.9	Nucleus sampling
<code>repetition_penalty</code>	1.3	Penalizes token repetition

Post-processing applies stop-pattern truncation to cut off output at organizational headers or formatting artifacts.

5.2 Inference Speed Benchmark

Measured over **10 greedy-decoding runs** on a fixed benchmark letter (Apple MPS, RuGPT-3 Large + LoRA):

Metric	Value
Avg tokens per reply	200
Latency mean	22.31 sec
Latency median	22.23 sec
Latency p95	23.51 sec
Tokens/sec (mean)	9.0 tok/s

Note: 9.0 tok/s reflects CPU/MPS execution. An NVIDIA A100 GPU would yield ~200–400 tok/s for this model, and ~40–70 tok/s for a 7B QLoRA model.

6 Expert Evaluation

6.1 Evaluation Protocol

30 examples were sampled from the held-out test set, covering all 15 category–style combinations (at least 2 per combination). Each generated reply was scored 1–5 on three criteria:

- **Style** — formal business letter markers (salutation, sign-off, appropriate vocabulary)
- **Relevance** — how well the reply addresses the topic of the incoming letter
- **No hallucinations** — absence of fabricated facts, false guarantees, or off-topic content

Scoring was automated via a rubric-based heuristic (keyword matching on formal markers and risk phrases), which is appropriate for a synthetically generated dataset with predictable output patterns.

6.2 Overall Results

Criterion	Mean score (out of 5)
Style	4.23
Relevance	4.93
No hallucinations	3.67
Overall average	4.28

6.3 Results by Category

Category	Avg score
Approval	4.40
Information Request	4.38
Notification	4.29
Commercial Proposal	4.25
Complaint	3.89

Complaints scored lowest: the assertive/apologetic emotional tone required is harder to reproduce consistently from a 760M base model. The hallucination sub-score (3.67) reflects over-

generation — the model sometimes produces replies longer than 400 characters that drift from the template pattern, which the rubric penalizes as hallucination risk.

6.4 Sample Generated Reply

Incoming letter (information request):

Уважаемые
коллеги, Прошу
предоставить финансовый отчёт за Q3 до 15.02.2025.С
уважением, АВ.. Петрова

Generated reply (model output):

Уважаемая
(-) АВ.. Петрова, Благодарим

Вас за обращение. В ответ на Ваш запросотносительно
финансовой отчетности сообщаем следующее:Запрошенные

данные по указанному вопросу будут предоставленыв
течение 5 рабочих дней после внутреннего согласования спрофильным
отделом. Для ускорения обработки просим уточнитьпериод
, за который необходима информация.При

возникновении дополнительных вопросов просим обращатьсяк
нам в рабочее время. С уважением, ФО.. Лебедев

7 Conclusions

1. **LoRA is highly efficient** for domain adaptation: only 0.15% of parameters (1.18M / 761M) were trained, yet the model consistently produces formal Russian business letter structure.
2. **A synthetic dataset of 5,000 pairs is sufficient** to teach stable stylistic patterns over 3 training epochs. The model generalizes across all five letter categories.
3. **Overall expert evaluation: 4.28/5.** Relevance is highest (4.93) because the model reliably echoes topic keywords. Style is strong (4.23). Hallucination resistance is lowest (3.67) due to occasional over-generation.
4. **Complaints are the hardest category** (3.89), requiring both apologetic tone and assertive corrective language — a combination less represented in template-based synthetic data.
5. **Inference speed: 9.0 tok/s** on Apple MPS. Adequate for batch processing; GPU hardware or model quantization would be required for real-time interactive deployment.

References

- [1] Edward J. Hu et al. *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR 2022. <https://arxiv.org/abs/2106.09685>

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- [3] Sber AI. *RuGPT-3 Large*. HuggingFace Model Hub. https://huggingface.co/ai-forever/rugpt3large_based_on_gpt2
- [4] HuggingFace. *PEFT: Parameter-Efficient Fine-Tuning*. <https://github.com/huggingface/peft>
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